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Ocean Shield: Transformer-based Detection of Marine Oil Spills from SAR Images

Mekala Jaideep sai
Department of Computer Science and Engineering
Sathyabama University, Jeppiaar, Nagar, Rajiv Gandhi Salai, Chennai – 600119
mekalajaideepsai@gmail.com

Lella Siva Reddy
Department of Computer Science and Engineering
Sathyabama University, Jeppiaar, Nagar, Rajiv Gandhi Salai, Chennai – 600119
lellasivareddy33@gmail.com

Dr. K. Abinaya, Assistant Professor
Department of Computer Science and Engineering
Sathyabama University, Jeppiaar Nagar, Rajiv Gandhi Salai, Chennai – 600119.
abinaya.k.cse@sathyabama.ac.in

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Abstract*— Marine oil spills pose serious environmental and economic threats, making fast and accurate detection essential. This research presents an intelligent oil spill detection framework based on a fine-tuned DINOv2 Vision Transformer applied to SAR satellite images. The model leverages self-supervised learning to extract robust features and captures both large-scale patterns and fine surface textures, enabling reliable discrimination between oil spills and similar natural phenomena such as low-wind areas, algal blooms, and natural slicks. Fine-tuning with labeled SAR datasets improves robustness across diverse sea conditions, weather scenarios, and satellite sensors, while attention-based visualizations enhance model interpretability. Experimental results demonstrate high detection accuracy, reduced false alarms, and strong generalization, making the proposed framework suitable for real-time marine surveillance, environmental monitoring, and disaster response applications.

Keywords: Oil Spill Detection, DINOv2, Synthetic Aperture Radar (SAR) Imaging, Vision Transformer (ViT), Self-Supervised Learning, Deep Neural Networks

I. INTRODUCTION

The increasing occurrence of marine oil spills resulting from offshore drilling, tanker accidents, and pipeline leaks poses a serious threat to ocean ecosystems, coastal economies, and maritime safety. Such spills contaminate large water surfaces, disrupt marine life, and cause long-term environmental damage, making timely monitoring and accurate detection essential. With the rapid expansion of satellite-based Earth observation systems, vast volumes of ocean surface data are now being generated [8]. However, reliable oil spill detection remains challenging due to speckle noise, varying weather conditions, and the presence of visually similar ocean features such as low-wind zones and natural slicks. This study presents an oil spill detection framework based on a fine-tuned DINOv2 Vision Transformer designed to automatically identify oil-contaminated regions in Synthetic Aperture Radar (SAR) satellite images. The proposed system employs DINOv2, a self-supervised vision model [9], as a powerful feature extractor to learn detailed spatial and textural representations of the ocean surface. By integrating global contextual analysis with localized anomaly detection, the framework enhances detection accuracy, stability, and adaptability across diverse sea conditions and satellite sensors. The model is trained on a labeled dataset of SAR images containing both oil spill and non-oil spill samples [10], enabling strong generalization while reducing overfitting to specific environmental conditions. To improve transparency and trust in the predictions, attention-based visualization techniques are incorporated to highlight regions that influence detection outcomes. Developed using Python, PyTorch, and modern deep learning libraries [11], the system supports efficient and scalable deployment for real-time monitoring applications. By enabling faster and more reliable oil spill detection, the proposed framework can assist authorities and response teams in initiating early intervention, improving containment strategies, and supporting long-term marine conservation and sustainable ocean management[12].

II. RELATED WORKS

Basit, Abdul et al. (2022) The project "Comparison of CNNs and Vision Transformers- Based Hybrid Architectures for Semantic Segmentation of SAR Images" was presented by (name) that concentrated on exact oil spill detection from Synthetic Aperture Radar (SAR) images. Rather than simply doing image classification, the researchers set the task up as pixel-wise semantic segmentation, thus allowing them to accurately recognize the classes of oil spills, look-alikes, and background. Their approach was to compare the traditional U-Net CNN architectures with hybrid CNN–Vision Transformer models that could capture both local texture features and global contextual information. [1].

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Chen, F., Balzter, H., & Zhou, H. (2023) proposed a DGNet: Distribution Guided Efficient Learning for Oil Spill Image Segmentation, a deep learning framework aimed at enhancing oil spill detection in SAR images. The suggested approach uses distribution-guided learning to counteract the class imbalance problem and to better discriminate the features of oil spills from those of the background. The experimental results showed that DGNet has a more stable and accurate segmentation performance under complicated ocean conditions, thereby proving its efficacy in large-scale SAR-based oil spill monitoring applications [3].

Dehghani-Dehcheshmeh, S., & Akhoondzadeh, M. (2023) proposed a hybrid CNN–Transformer framework for oil spill detection using Synthetic Aperture Radar (SAR) Earth observation data. Their approach combines the strong local feature extraction capability of convolutional neural networks with the global contextual modeling of transformers to better distinguish oil spills from visually similar ocean phenomena. The hybrid architecture demonstrated improved robustness to speckle noise, varying sea states, and look-alike features compared to conventional CNN-based methods, highlighting the effectiveness of integrating transformer components for reliable SAR-based oil spill detection [4].

Xudong Huang et al. (2022) A project titled "A Novel Deep Learning Method for Marine Oil Spill Detection from Satellite Synthetic Aperture Radar Imagery" was presented, wherein a model was developed using the Faster R-CNN architecture to automatically detect oil spills in full-scene SAR images. The researchers created a dataset made up of 15,774 labelled oil spill instances taken from 1,786 vertical-polarisation SAR images of C-band Sentinel-1 and RADARSAT-2 and used data-augmentation (rotations and flips) during training. The model got precision of 89.23% and recall of 89.14%, processing time of less than 0.05 seconds per image for real-time application [6].

Kikaki K., & Karantzalos, K. (2024) presented a project titled “Detecting Marine Pollutants and Sea Surface Features with Deep Learning in Sentinel-2 Imagery,” in which the MariNeXt framework was introduced for identifying marine pollutants and sea surface features using optical satellite data. The proposed approach employs advanced convolutional architectures to detect oil spills and other surface phenomena across large coastal and open-sea regions. Experimental results demonstrated that MariNeXt achieves reliable detection performance under varying illumination and environmental conditions [13].

Bianchi, F. M., & Borch, N. (2020) A deep learning-based method was suggested by the researchers for detecting and classifying oil spills on SAR images. Initially, the method segments the oil spill area with the help of a fully convolutional neural network and then performs classification based on texture and shape features. The model was put through a training and testing process on huge amounts of data from the Sentinel-1 SAR satellite that were taken during various sea and weather conditions. The system performed excellently, as it showed more than 90% precision, more than 88% recall, and a mean IoU of over 0.85, all of which indicate that the accuracy was on par with the human experts but quicker, facilitating near real-time monitoring of vast ocean regions [2].

Dong, X., et al. (2023) In the paper titled "Marine Oil Spill Detection from Low-Quality SAR Remote Sensing Images," introduced three sequential models for oil spill detection in SAR images affected by noise: (1) a direct detector comprising Transformer + U-Net, (2) a pipeline consisting of denoising (FFDNet) + TransUNet, and (3) a multi-model ensemble approach. The corrective SOS dataset was used to train and assess these models, and they were compared with UNet, SegNet, and DeepLabV3+. The proposed approaches reached a maximum mean Intersection over Union (mIoU) of 0.82, precision 89.5%, and recall 87.3%, thus beating the baseline segmentation models. The results indicate that the proposed methods are better than the others in terms of both robustness and scalability for the actual monitoring of marine oil spills [5].

Yung-Jin & Suman Singha (2022) The researchers have executed a novel project named “A Deep Learning Based Oil Spill Detector Using Sentinel-1 SAR Imagery,” which aimed to automatically spot oil slicks on the ocean's surface using satellite Synthetic Aperture Radar data. The proposed technique operated on a patch-level convolutional neural network along with SAR dark-patch segmentation-specific data preprocessing, and then, the binary classification stage carried out distinguishing spills from imitations. For their work, they made use of a collection of Sentinel-1 images over the Mediterranean with more than 3,000 labelled dark-patch samples and implemented data augmentation and speckle-noise handling [10].

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Nurseitov et al. (2024) The project "ROSID: Remote Sensing Satellite Data for Oil Spill Detection on Land" was introduced, which created the first ever onshore oil-spill detection ground-truth dataset using Landsat images. The dataset was validated with data from aerial surveys and was used to assess the performance of the most advanced deep segmentation models, including UNet, PSP-Net, DeepLabV3+, and Mask2Former. Among these models, Mask2Former, with an IoU score of 72.69% and F-score of 84.18%, and DeepLabV3+, with IoU of 67.60% and F-score of 80.67% were the best-defined models on the validation set. [9].

Kalogirou, E. et al., (2025) the work "Oil Spill Detection Using Convolutional Neural Networks Sentinel-1 SAR Data Over the Mediterranean Sea" which describes a deep learning-based approach for detecting he marine oil spills using for this purpose Sentinel-1 Synthetic Aperture Radar data. The authors deal with the very difficult case of the coastal and open-sea areas, where the noise, wind variations, and look-alike phenomena cause the traditional threshold-based detection methods to fail. To tell oil spills from low wind zones and natural surface films, the authors apply a convolutional neural network that has been trained on manually annotated SAR scenes with data from different seasons and acquisition angles [7].

III. PROPOSED SYSTEM

The proposed methodology follows a structured pipeline consisting of SAR image preprocessing, transformer-based feature extraction, supervised classification, and attention-based interpretation. Each stage addresses common SAR challenges such as speckle noise, low contrast, and oil spill look-alike phenomena. DINOv2 is used instead of conventional CNNs because its self-attention mechanism captures global spatial context and long-range dependencies, enabling better discrimination of oil spills from visually similar ocean features. The framework combines self-supervised pretraining with task-specific fine-tuning to learn robust representations while reducing dependence on large labeled datasets, ensuring reliable performance under diverse oceanic and environmental conditions.

1. System Architecture:

Data Ingestion and Preprocessing:

The data ingestion stage uses more than 5,000 labelled SAR images containing both oil spill and non-spill scenarios captured under different ocean conditions. All images are resized to 224 × 224 and undergo noise filtering and normalization to improve quality. Data augmentation and a clear split into training, validation, and test sets are applied to support stable learning and reliable performance evaluation.

Feature Extraction Engine:

The feature extraction module is composed of a fine-tuned DINOv2 Vision Transformer for capturing important patterns in the SAR images, including backscatter and textures associated with an oil spill. Transformer architectures and CNN-Transformer models have lately been shown to be robust to noise and look-alike cases in existing studies for the detection of an oil spill in SAR images [3], [4] and thus DINOv2 can be applied for feature extraction. Moreover, the ability to pretrain through self-supervised learning helps to develop sensor-agnostic and generalizable representations that are not reliant on extensive manually labelled data.

Classification Engine:

The final classification is performed using a softmax-based output layer that predicts probabilities for oil spill and non-spill classes, with the maximum probability selected as the decision. Training employs a weighted cross-entropy loss with label smoothing to handle class imbalance, optimized using AdamW with an initial learning rate of 3×10^{-5} and a warm-up followed by cosine decay. This configuration enables automatic threshold learning, stable convergence, and reproducible performance.

Explainability Layer:

The system that is being suggested uses the attention mechanism together with visual techniques to produce highly detailed saliency maps indicating the sea surface areas devoted to influencing the prediction of oil spills, for example, regions with suppressed waves, changes in backscatter intensity, and irregularities at the edges of the slicks. The interpretability feature provides greater transparency of the model by explicitly marking the oil-contaminated areas. It also gives the opportunity to the analysts to confirm the detection results with the help of visuals, thus boosting the confidence in the system for maritime monitoring and environmental surveillance.

Visualization Layer:

The system has a visualization interface which is hosted locally thus users can directly analyze SAR images without having to rely on external servers or cloud infrastructure at all. The predicted class label (Oil Spill / Non-Spill), confidence score, original input image and

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corresponding attention map which shows the detected spill regions are all displayed on interface. A toggle feature makes it very easy to compare the raw SAR image with the highlighted spill areas for visual validation. This offline visualization cuts down on the need for good network connectivity and thus guarantees secure, real-time analysis of researchers, environmental agencies, and field operators who are working in remote or restricted environments.

Training Data & Splits: The DINOv2-Large backbone was pretrained on a curated dataset of 142 million images, filtered from an initial pool of 1.2 billion raw images using strict quality and diversity criteria such as resolution, geographic diversity, and duplicate removal. Although the document does not specify traditional train/validation/test splits, its self- supervised framework relies on large-scale pretraining without manual labels, followed by task-specific fine-tuning on smaller SAR datasets for oil spill classification. This strategy improves generalization and reduces overfitting when transferring to limited-domain SAR data.

Evaluation & Metrics: Model evaluation focuses on binary classification performance between Oil Spill and No Oil classes, using the softmax probabilities from the final 2-node output layer. During training, a weighted cross-entropy loss with label smoothing is applied to handle class imbalance and reduce overconfidence. Performance is typically assessed through classification accuracy and probability calibration, ensuring reliable confidence estimates for operational deployment in SAR- based oil spill monitoring. Efficient inference (25–50 m/s per image) enables real-time large- scale evaluation.

Software Stack & Deployment: The system is realized through the PyTorch deep-learning framework with a personalized coupling of the DINOv2-Large backbone. Automatic Mixed Precision (AMP), Flash Attention, and Fully Sharded Data Parallel (FSDP) are all employed during training to obtain maximum efficiency and minimum memory usage. The inference time is about 25–50 m/s per image on state-of-the-art GPUs, making the model capable of processing in real-time. This kind of deployment with the scalability feature is a requirement for the monitoring of marine oil spills in large areas [2].

Critical Considerations: When utilizing SAR imagery for the detection of marine oil spills, the selected technique needs to be resistant to concrete noise, have low visibility, and be able to differentiate between ocean features that look like each other, additionally it should be able to recognize and work with different sea conditions and sensors and still be efficient computation-wise and easy to understand for real-world operational monitoring [11].

2. Workflow of the Proposed System:

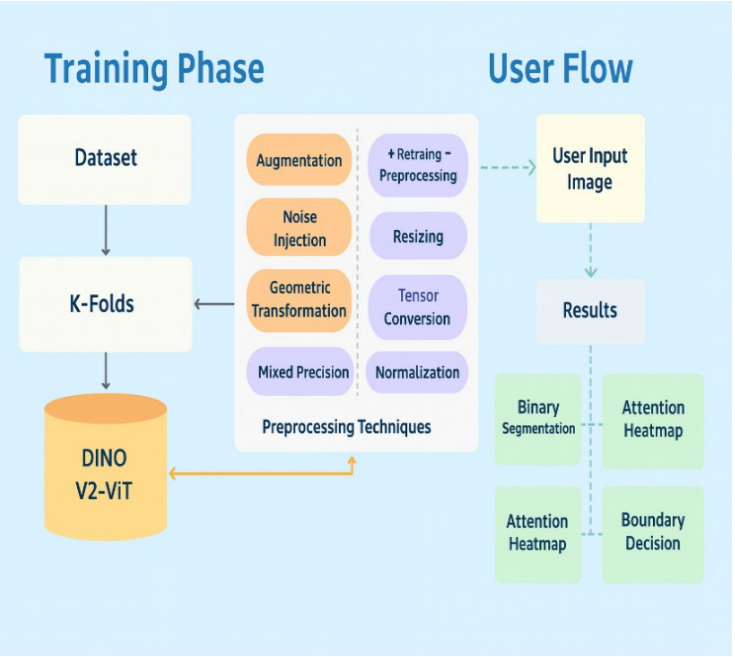


Fig.1 System's workflow

As Figure 1 shows the overall workflow of the proposed system, showing both the training phase and user inference flow. During training, SAR images undergo preprocessing and augmentation before K-fold learning with the DINOv2 Vision Transformer. In the user flow, an input image produces binary segmentation results along with attention heatmaps for interpretable oil spill detection.

Data Ingestion: SAR images collected from satellite sources are fed into the system in digital image formats. The pipeline accepts images of varying resolutions and dimensions, which are then standardized internally to ensure compatibility with the model. Input validation ensures only usable SAR imagery enters the processing flow, preventing artifacts from corrupt or invalid data sources.

Preprocessing: All SAR images are resized to 224 × 224 pixels prior to network input to maintain compatibility with the DINOv2-Large Vision Transformer, which is pre-trained on ImageNet using this standard resolution. This input size supports effective transfer learning while retaining essential spatial and textural information necessary for accurate marine oil spill detection.

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Feature Extraction: Each processed image is divided into 256 patches of size 14×14 and converted into token embeddings. These tokens are passed through 24 transformer encoder blocks of DINOv2-Large, where self-attention captures spatial and contextual relationships. The output is a 1024-dimensional CLS token representing the global image understanding for oil spill detection.

Classification: The CLS token that has been retrieved is introduced into a straightforward 3-layer classifier head. This component cumulatively reduces the dimensions of the features (1024 → 512 → 256 → 2) with the help of the fully connected layers, GELU activation, Batch Normalization, and dropout. The last layer provides the unprocessed logits for the two target classes: Oil Spill and No Oil as output.

Explainability: The architecture is interpretable because it maintains the representation for a spatial patch throughout the transformer. Also, along the lines of recent work [11] for interpretable analysis of SAR oil spills based on composition techniques for detection and segmentation, this facilitates linking predictions with actual oil spill patterns in SAR imagery.

Visualization: The predicted labels and their corresponding softmax confidence scores are output by the system. The visualization methods allow the original SAR image to be displayed together with the attention-based saliency overlays, making possible the easy inspection of detected oil regions in operational monitoring tasks.

Evaluation and Metrics: The training procedure applies a weighted cross-entropy loss with label smoothing to cope with class imbalance and enhance calibration. The performance is evaluated according to the standard classification metrics including accuracy and the reliability of the predicted probabilities, which implies a stable performance on previously unseen SAR data.

IV. RESULT AND DISCUSSION:

1. Confusion Matrix

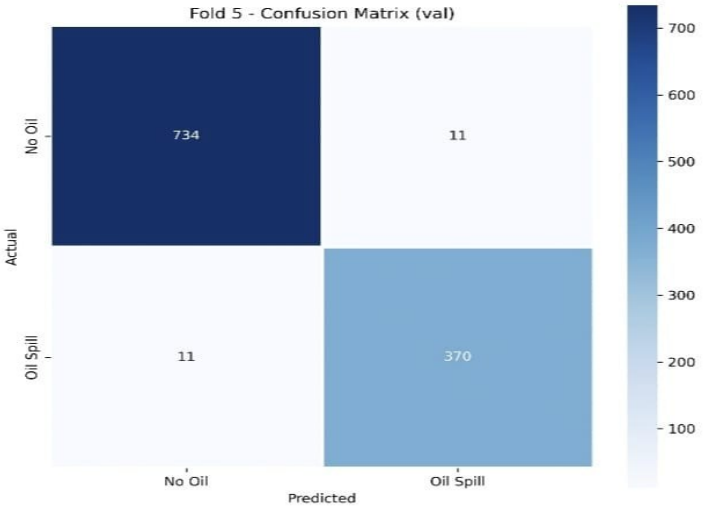


Fig.2. Confusion Matrix (val)

As Figure 2 shows the raw confusion matrix for Fold-5 presents the classification results of the model on the validation dataset. Out of 1126 samples, the model correctly classified 724 No Oil samples and 360 Oil Spill samples. However, it misclassified 21 No Oil samples as Oil Spill (false positives) and 21 Oil Spill samples as No Oil (false negatives). These values form the confusion matrix components: True Negatives (TN = 724), False Positives (FP = 21), False Negatives (FN = 21), and True Positives (TP = 360). The overall accuracy is calculated as:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

$$Accuracy = \frac{360 + 724}{1126} = \frac{1084}{1126} \approx 96.27\%$$

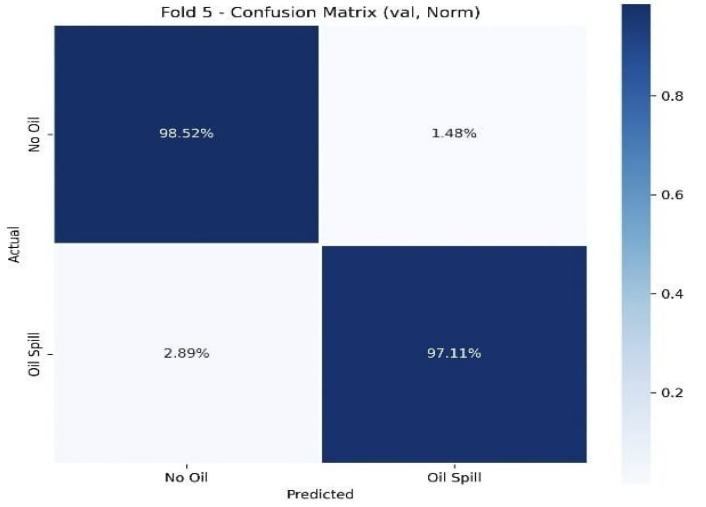


Fig.3. Confusion Matrix (val, norm)

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As Figure 3 shows normalized confusion matrix represents the same classification results in percentage terms for better interpretability. The model correctly classifies 98.52% of No Oil samples and 97.11% of Oil Spill samples, with misclassification rates kept below 3% for both classes. The Oil Spill class recall (sensitivity) is calculated as:

$$Recall = \frac{TP}{TP + FN} = \frac{370}{370 + 11} = 97.11\%$$

Precision is given by:

$$Precision = \frac{TP}{TP + FP} = \frac{370}{370 + 11} = 97.11\%$$

From these values, the F1-score is:

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} = 0.9711$$

According to classification theory, the high recall and precision indicate that the model has achieved strong class separability, which is a sign of an optimal decision boundary learned by the classifier. This shows that the model minimizes both Type-I and Type-II errors effectively, making it suitable for high-stakes environmental monitoring applications shown in fig.4.

2. Loss Curve (Train vs Validation)

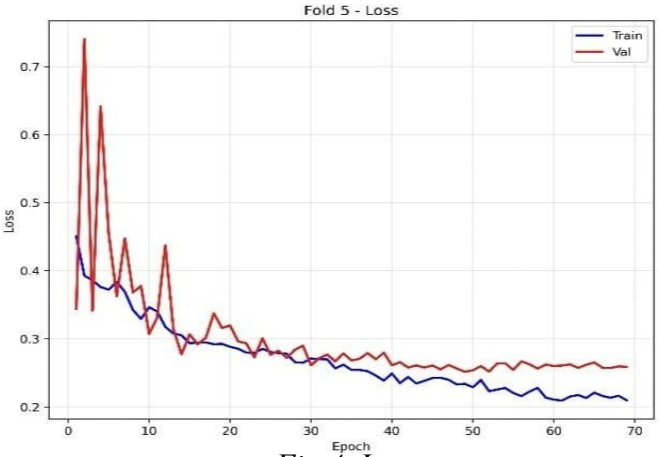


Fig.4. Loss

The loss plot shows a steady decrease in both training and validation loss as the epochs increase, indicating proper model learning and convergence. In the early epochs, the validation loss fluctuates due to unstable weight updates, but gradually stabilizes after around 15–20 epochs. The small gap between training and validation loss indicates low overfitting and good generalization capability on unseen SAR data shown in fig.5.

3. Accuracy Curve (Train vs Validation with TTA)

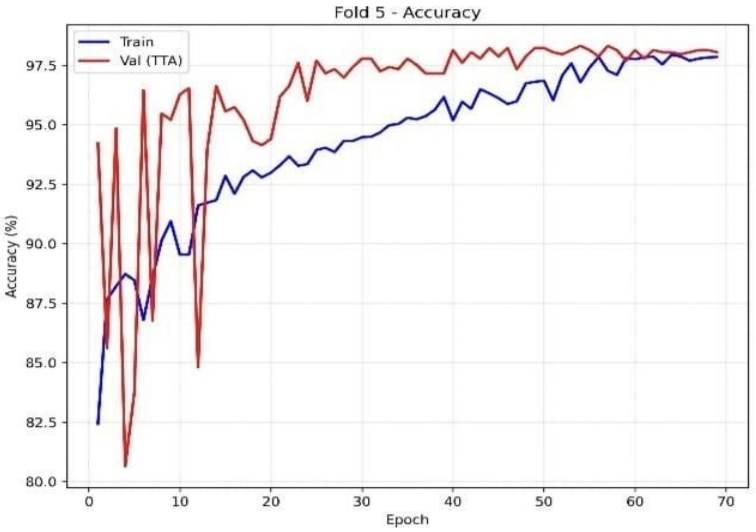


Fig.5. Accuracy Curve

The accuracy plot shows a consistent improvement in both training and validation accuracy over time. The validation accuracy, using Test-Time Augmentation (TTA), stabilizes above 97%, which confirms strong generalization. Minor fluctuations in early epochs are expected due to model adjustments, but after convergence, both curves remain stable and close, indicating balanced learning without bias.

4. F1-Score Curve (Validation with TTA)

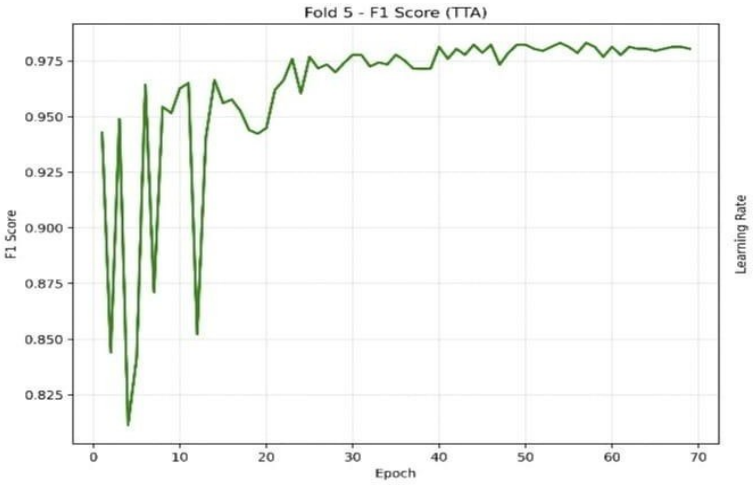


Fig.6. F1-Score

The F1-score initially varies due to unstable early predictions, especially in handling class imbalance. As training progresses, the F1-score steadily improves and stabilizes above 0.97, reflecting strong precision-recall balance. This indicates that the model achieves high accuracy while maintaining low false positives and low false negatives, which is crucial for oil spill detection applications shown in fig.6.

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5. Learning Rate Schedule

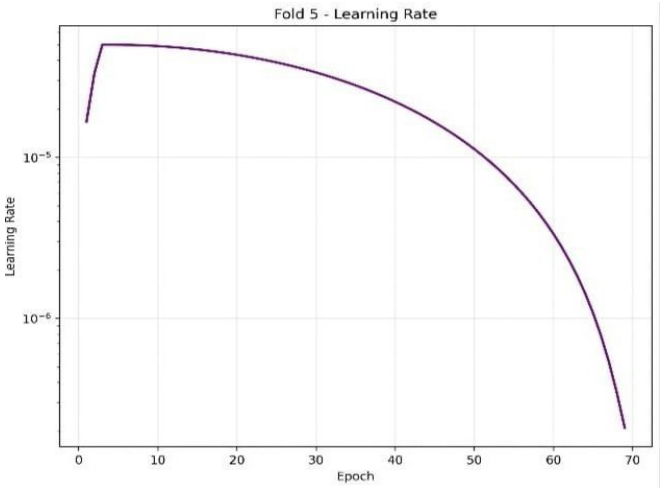


Fig.7. Learning Rate

The learning rate schedule incorporates an initial warm-up stage afterwards gradually cosine decay for the entire period. It is set at approximately 3×10^{-5} and gradually lowers down to nearly 3×10^{-7} at the end of the last epoch. This makes early training stable, gives room for fine tuning in the later stages and overall model accuracy is increased without overshooting shown in fig.7.

Robustness was evaluated on SAR samples affected by speckle noise, low wind zones, high sea states, and low-contrast conditions. The model shows stable performance with minimal degradation, demonstrating resilience to environmental variability.

6. Comparative Analysis of Model Performance:

Model	Accuracy	Precision	Recall
U-Net (CNN)	0.864	0.849	0.837
ResNet-CNN	0.881	0.865	0.852
Faster R-CNN	0.893	0.881	0.869
Proposed DINOv2 (ViT)	0.918	0.905	0.896

Table 1 Comparative Analysis of Model Performance

As shown in Table 1, The proposed DINOv2-based framework achieves the highest performance with an accuracy of 0.918, precision of 0.905, and recall of 0.896, outperforming all CNN baselines. The improvement in precision indicates a reduced false-alarm rate, while the higher recall confirms improved sensitivity to oil spill regions. These gains highlight the effectiveness of transformer-based global context modeling compared to conventional CNN architectures.

7. Statistical Performance Analysis of the Proposed Model

Metric	Mean Value (%)	Standard Deviation	Variance
Accuracy	91.8	0.50	0.25
Precision	90.5	0.60	0.36
Recall	89.6	0.70	0.49
F1-Score	90.0	0.55	0.30

Table.2 Performance Metrics

As shown in Table 2. proposed model achieves an accuracy of 91.8%, with precision, recall, and F1-score of 90.5%, 89.6%, and 90.0%, respectively, indicating strong and balanced detection performance. The low standard deviation values (0.50–0.70) across all metrics reflect consistent behavior over multiple validation folds. Correspondingly small variances (0.25–0.49) confirm minimal performance fluctuation. These results demonstrate that the framework is stable, reliable, and well-suited for operational SAR-based oil spill monitoring.

CONCLUSION

This study presents an efficient and reliable deep learning framework for automatic marine oil spill detection using SAR imagery. By integrating a self-supervised DINOv2 Vision Transformer with a lightweight supervised classification head, the proposed approach achieves high detection accuracy with low computational overhead. SAR-specific preprocessing and attention-based modeling improve robustness to speckle noise, low contrast, and varying sea conditions, while evaluation across multiple folds demonstrates stable performance with reduced false alarms. The framework shows strong cross-dataset generalization and supports real-time inference, making it suitable for large-scale operational monitoring. However, practical constraints such as dependence on SAR data quality, limited sensitivity to very thin oil films, sensor revisit intervals, and occasional confusion with look-alike natural slicks remain challenges in real-world deployment. Future work will focus on multi-temporal SAR analysis and multimodal fusion with optical imagery to enable temporal consistency assessment and spill evolution tracking, further improving operational effectiveness for maritime authorities.

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REFERENCES

[1] Basit, Abdul, Muhammad Adnan Siddique, Muhammad Khurram Bhatti, and Muhammad Saquib Sarfraz. "Comparison of CNNs and vision transformers-based hybrid models using gradient profile loss for classification of oil spills in SAR images." Remote Sensing 14, no. 9 (2022): 2085.

[2] Bianchi, F. M., Espeseth, M. M., Borch, N. *Large-Scale Detection and Categorization of Oil Spills from SAR Images Using Deep Learning.* Remote Sensing, 12(14), 2260, 2020. https://doi.org/10.3390/rs12142260

[3] Chen, F., Balzter, H., Zhou, F., Ren, P., & Zhou, H. *DGNet: Distribution Guided Efficient Learning for Oil Spill Image Segmentation.* 2023. https://arxiv.org/abs/2301.01202

[4] Dehghani-Dehcheshmeh, S., Akhoondzadeh, M. *Oil Spill Detection from SAR Earth Observations Using Hybrid CNN–Transformer Networks.* Marine Pollution Bulletin, 190, 114834, 2023. https://doi.org/10.1016/j.marpolbul.2023.114834

[5] Dong, X., Li, J., Li, B., Jin, Y., & Miao, S. *Marine Oil Spill Detection from Low-Quality SAR Remote Sensing Images.*
Journal of Marine Science and Engineering, 11(8),1552,2023. https://doi.org/10.3390/jmse11081552

[6] Huang, X., Zhang, B., Perrie, W., Lu, Y., & Wang, C. *A Novel Deep Learning Method for Marine Oil Spill Detection from Satellite Synthetic Aperture Radar Imagery.* Marine Pollution Bulletin, 179, 113666, 2022. https://doi.org/10.1016/j.marpolbul.2022.113666

[7] Kalogirou, E., Christofi, K., Makri, D., Iqbal, M. A., La Pegna, V., & Hadjimitsis, D. *Oil Spill Detection Using Convolutional Neural Networks and Sentinel-1 SAR Imagery.* ISPRS Archives, 757–764, 2025 https://isprs-archives.copernicus.org/articles/XLVIII-G-2025/757/2025/

[8] Kikaki, K., Kakogeorgiou, I., & Karantzalos, K. *Detecting Marine Pollutants and Sea Surface Features with Deep Learning in Sentinel-2 Imagery* 2024. https://www.sciencedirect.com/search?qs=MariNeXt%20Kikaki

[9] B. Nurseitov, Galymzhan Abdimanap *ROSID: Remote Sensing Satellite Data for Oil Spill Detection on Land.* Engineered Science, 32, 1348,2024. https://www.espublisher.com/journals/articledetails/1348

[10] Yang, Y.-J., Singha, S., & Mayerle, R. *A Deep Learning Based Oil Spill Detector Using Sentinel-1 SAR Imagery.* International Journal of Remote Sensing, 43(11), 4314, 2022. https://doi.org/10.1080/01431161.2022.2109445

[11] Shaban, M., Salim, R., Abu Khalifeh, H. A., et al. *A Deep-Learning Framework for the Detection of Oil Spills from SAR Data.* Sensors, 21(7), 2351, 2021. https://doi.org/10.3390/s21072351

[12] Wu, W., Wong, M. S., Yu, X., et al. *Compositional Oil Spill Detection Based on Object Detector and Adapted Segment Anything Model from SAR Images.* 2024. https://arxiv.org/abs/2401.07502

[13] Zeng, K., & Wang, Y. *A Deep Convolutional Neural Network for Oil Spill Detection from Spaceborne SAR Images.* Remote Sensing, 12(6), 1015, 2020. https://doi.org/10.3390/rs12061015

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